Title (Units): **COMP7045 Natural Language Processing and Large Language Models (3,2,1)**

Course Aims: To give students a solid background of natural language processing and Large Language Models. To learn creative designs and implementations, including critical thinking of methodologies, problem analytics, practical techniques and tools for a wide range of downstream applications. Students after taking this course will be able to: 1) identify and creatively apply advanced techniques of natural language processing to solve downstream problems, and 2) build modules to creatively design natural language systems, and critically analyze their effectiveness; and 3) gain advanced knowledge of Large Language Models, with the ability to design, implement, and evaluate solutions for specialized tasks.

Prerequisite: Basic knowledge in artificial intelligence

Course Intended Learning Outcomes (CILOs):
Upon successful completion of this course, students should be able to:

No.	Course Intended Learning Outcomes (CILOs)
	Knowledge
1	Describe the fundamental concepts and methodologies of natural language processing and large language models
2	Explain the advantages and limitations of methods developed for different scenarios
3	Identify relevant language processing techniques to meet real-world needs
	Professional Skill
4	Apply specific methods and techniques in a wide-range of downstream applications
5	Design and evaluate the solutions to technical problems
	Attitude
6	Engage in life-long independent learning for problem-solving and advance state-of-the art solutions

Calendar Description: This course is to provide an in-depth knowledge of natural language processing and large language models. Topics include: fundamental concepts, conceptual modeling of a natural language, deep learning techniques for focused problems, large language models, and critical thinking of downstream applications. The students will have a thorough understanding of natural language processing technique and principles of large language models, and be able to creatively design models and critically analysis their effectiveness after taking this course.

Teaching and Learning Activities (TLAs):

CILOs	Type of TLA
1-5	Students will learn essential knowledge of natural language processing and large language models through lectures and tutorials. There will be written assignments, quizzes and final examination to evaluate the students' level of understanding.
4-5	Students will learn critical algorithms and techniques of some downstream tasks and applications through lectures and tutorials. Laboratory sessions will also be designed so that students could apply what they have learnt in lectures. There will be laboratory exercises and quizzes.
4-6	Students are required to conduct a project based on a selected topic about NLP and large language models and give a formal presentation on their proposed method. Peer evaluation by Instructor(s) and teaching assistant(s) will be involved.

Assessment:

No.	Assessment Methods	Weighting	CILOs to be addressed	Description of Assessment Tasks
1	Continuous Assessment – assignments	25%	1-6	Assignments, quizzes and labs will be used to consolidate their knowledge and develop their skills

	and quizzes/tests			in natural language processing and large language models.
2	Continuous Assessment – project	25%	1-6	Individual/group project and report will further strength their understanding and problem solving skills. Peer evaluation will be involved.
3	Examination	50%	1-6	Examination will be used to assess students' overall understanding in the concepts, algorithms, methodologies and their ability in applying these knowledge to solve problems.

Assessment Rubrics:

Excellent (A)	• Achieve all CILOs, demonstrating a good mastery of both the theoretical and practical aspects of the knowledge and skills associated with natural language processing and large language models • Able to develop correct solutions to problems, accompanied by critical thinking, analytical thinking and creative thinking • Demonstrate a thorough understanding and solid knowledge of natural language processing and large language models • Able to apply a variety of techniques and relevant knowledge for fulfilling the real-world needs
Good (B)	• Achieve the first five CILOs, demonstrating a good understanding of the concepts and underlying methodologies • Able to develop correct solutions to problems, accompanied by adequate explanations • Demonstrate a competent level of knowledge of natural language processing and large language models • Able to make use of appropriate techniques and knowledge and apply them to familiar situations and problems
Satisfactory (C)	• Achieve most of the first five CILOs, demonstrating a basic level of understanding of the concepts and underlying methodologies • Able to provide acceptable solutions to problems • Demonstrate an adequate level of knowledge of natural language processing and large language models • Able to make use of some techniques and knowledge and apply them to familiar situations
Fail (F)	• Achieve less than three of the CILOs, with little understanding of the associated concepts and underlying methodologies • Unable to provide solutions to simple problems • Knowledge of natural language processing and large language models falling below the basic minimum level • Unable to apply techniques and knowledge to situations or problems

Course Content and CILOs Mapping:

Content		**CILO No.**
I	Introduction to Natural Language Processing (NLP) and Core Concepts	1
II	Text Preprocessing Techniques	1,2
III	Statistical Language Models and Syntactic Analysis	1,2,3
IV	Word Representation and Deep Neural Networks	4,5
V	Transformers and Neural Language Models	1,2,3
VI	Training Large Language Models (LLMs) and Foundation Models	5,6
VII	Downstream Applications of NLP and LLMs Techniques	4,5,6

References:

- Daniel Jurafsky and James H. Martin. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition with Language Models, 3rd edition. Prentice Hall. 2025.
- Christopher D. Manning and Hinrich Schütze. Foundations of Statistical Natural Language Processing. The MIT Press. 1999.
- Jason Brownlee. Deep Learning for Natural Language Processing. Machine Learning Mastery, 2018.
- Steven Bird, Ewan Klein, and Edward Loper. Natural Language Processing with Python. 1st edition, O'Reilly Media; 2009
- Lane H and Dyshel M. Natural Language Processing in Action. Simon and Schuster. 2025.
- Raschka S. Build a Large Language Model (From Scratch). Simon and Schuster. 2024.

Course Content:

<u>Topic</u>

I. Introduction to Natural Language Processing (NLP) and Core Concepts

II. Text Preprocessing Techniques
   A. Text Preprocessing: Tokenizing, Segmentation, Lemmatization, Stemming, etc
   B. Regular expression

III. Statistical Language Models and Syntactic Analysis
   A. N-grams Language Models
   B. Smoothing and Evaluation
   C. Vector Semantics and Vector Space Model
   D. Part-Of-Speech Tagging and Syntactic Parsing

IV. Word Representation and Deep Neural Networks
   A. Word Embedding
   B. Recurrent Neural Networks and Convolutional Neural Networks
   C. Recursive Neural Networks
   D. Network Embedding

V. Transformers and Neural Language Models
   A. Attention in Sequence-to-Sequence Models
   B. Introduction to Transformer
   C. Neural Language Models

VI. Training Large Language Models (LLMs) and Foundation Models
   A. Pre-training and Fine-tuning
   B. Model Alignment, Prompting, and In-Context Learning
   C. Overview of Foundation Models (GPT, LLaMA, DeepSeek and others)

VII. Downstream Applications of NLP and LLMs Techniques
   A. Sentiment Classification
   B. Machine Translation
   C. Question Answering and Chatbots
   D. Retrieval Augmented Generation
   E. Text Summarization